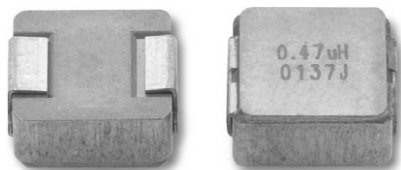


IHLP® Commercial Inductors, Low DCR Series



LINKS TO ADDITIONAL RESOURCES



RoHS
COMPLIANT
HALOGEN
FREE

FEATURES

- Shielded construction
- Lowest DCR/µH, in this package size
- Handles high transient current spikes without saturation
- Ultra low buzz noise, due to composite construction
- Excellent temperature stability for inductance and saturation
- Excellent DC/DC energy storage up to 1 MHz to 2 MHz. Filter inductor applications up to SRF (see “Standard Electrical Specifications” table)
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

APPLICATIONS

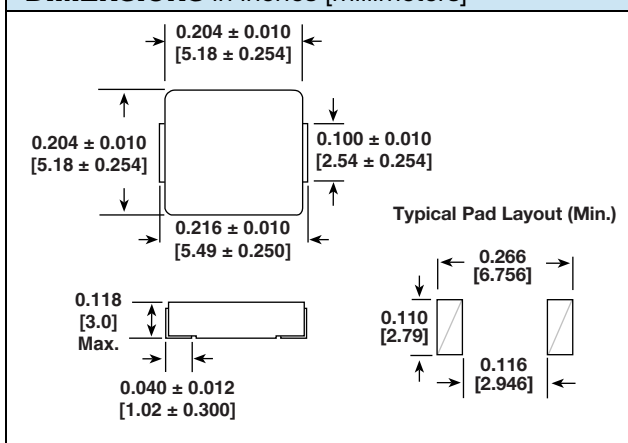
- PDA / notebook / desktop / server applications
- High current POL converters
- Low profile, high current power supplies
- Battery powered devices
- DC/DC converters in distributed power systems
- DC/DC converter for field programmable gate array (FPGA)

STANDARD ELECTRICAL SPECIFICATIONS						
L ₀ INDUCTANCE ± 20 % AT 100 kHz, 0.25 V, 0 A (µH)	DCR TYP. 25 °C (mΩ)	DCR MAX. 25 °C (mΩ)	HEAT RATING CURRENT DC TYP. (A) ⁽¹⁾	SATURATION CURRENT DC TYP.		SRF TYP. (MHz)
				(A) ⁽²⁾	(A) ⁽³⁾	
0.1	2.6	2.9	26.3	20.5	31.7	312
0.22	3.5	3.9	20.0	8.8	13.4	144
0.33	4.5	5.0	18.5	8.7	13.0	121
0.47	5.4	6.0	15.4	7.4	11.0	89
1	10.0	11.0	11.6	6.4	9.4	62
1.5	17.1	18.5	9.0	5.2	7.6	46
2.2	22.5	25.0	8.5	4.9	7.2	35
3.3	36.4	40.4	6.3	4.7	6.8	30
4.7	54.0	60.0	5.1	3.7	5.4	25
5.6	63.0	70.6	4.2	3.0	4.4	21
10	122.1	131.9	3.2	1.4	2.1	16
22	260.0	270.0	2.12	1.13	1.61	10.9

Notes

- All test data is referenced to 25 °C ambient
- Operating temperature range -55 °C to +125 °C
- The part temperature (ambient + temp. rise) should not exceed 125 °C under worst case operating conditions. Circuit design, component placement, PWB trace size and thickness, airflow and other cooling provisions all affect the part temperature. Part temperature should be verified in the end application
- Rated operating voltage (across inductor) = 40 V
- (1) DC current (A) that will cause an approximate ΔT of 40 °C
- (2) DC current (A) that will cause L₀ to drop approximately 20 %
- (3) DC current (A) that will cause L₀ to drop approximately 30 %

DIMENSIONS in inches [millimeters]



DESCRIPTION

IHLP-2020CZ-11	4.7 µH	± 20 %	EK	e3
MODEL	INDUCTANCE VALUE	INDUCTANCE TOLERANCE	PACKAGE CODE	JEDEC® LEAD (Pb)-FREE STANDARD

GLOBAL PART NUMBER

I	H	L	P	2	0	2	0	C	Z	E	K	4	R	7	M	1	1
PRODUCT FAMILY				SIZE						PACKAGE CODE		INDUCTANCE VALUE			TOL.	SERIES	

PACKAGE CODE OPTIONS

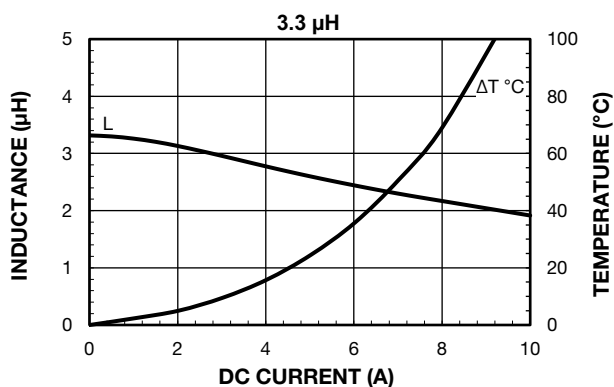
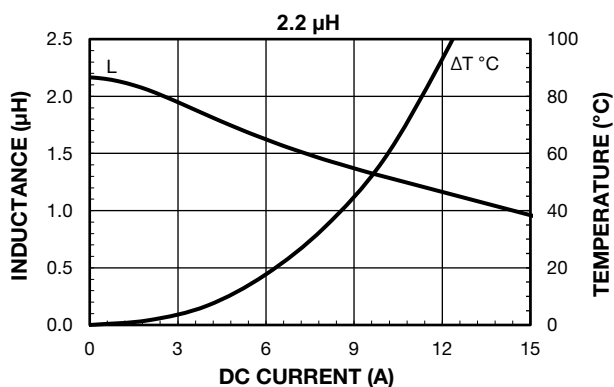
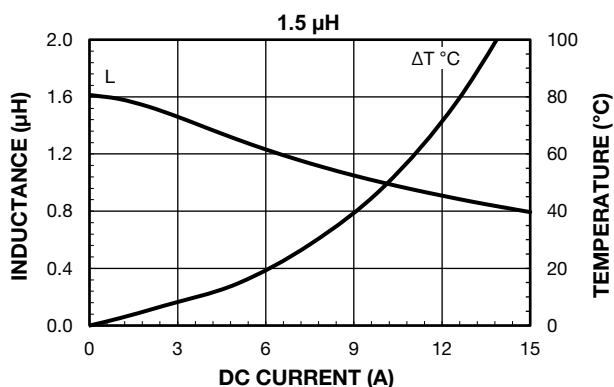
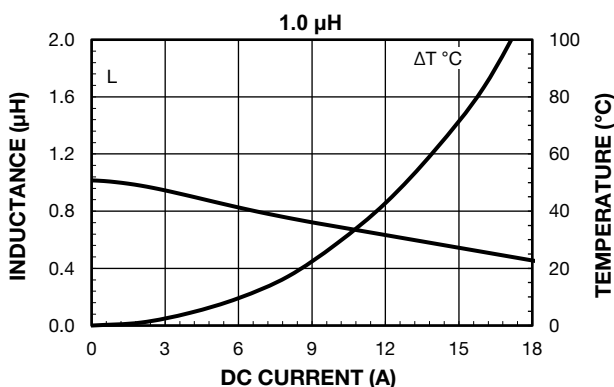
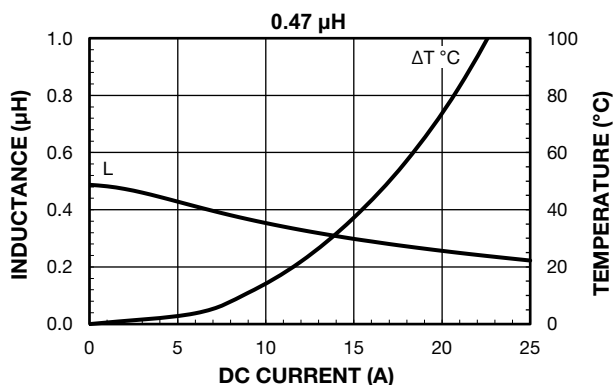
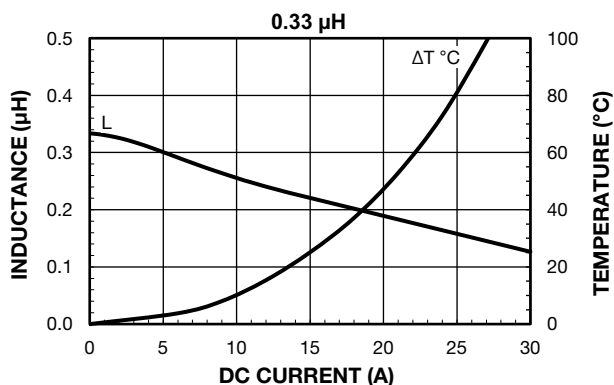
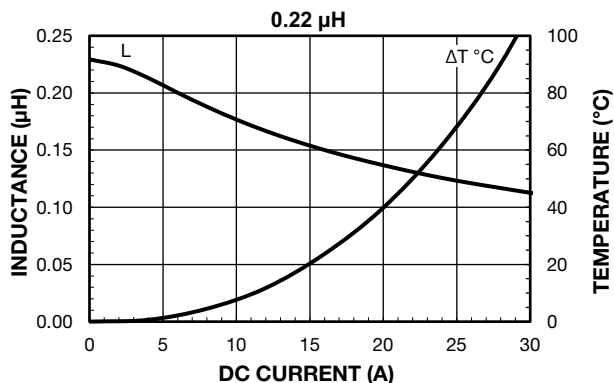
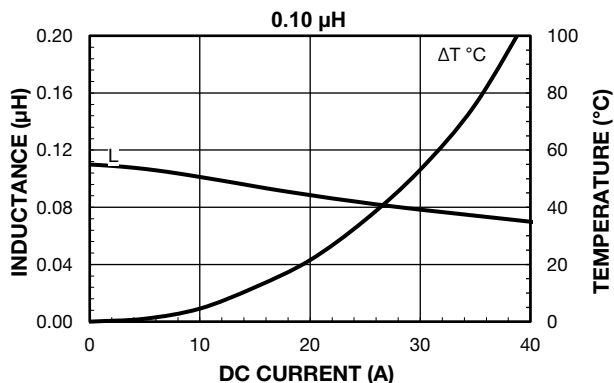
- EK** = tape and reel packaging (2500 pcs on 13-inch reel)
- ER** = tape and reel packaging (2000 pcs on 13-inch reel)

Note

- For additional packaging details see “[Packaging Methods](#)”

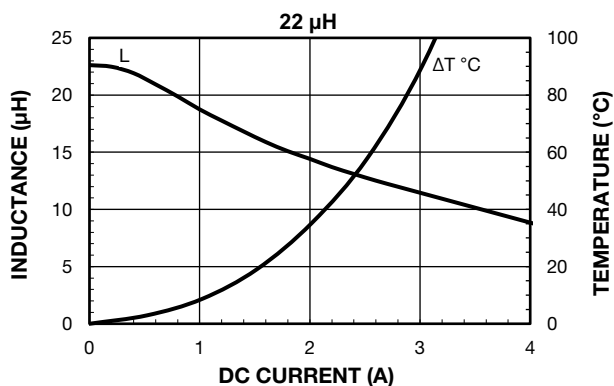
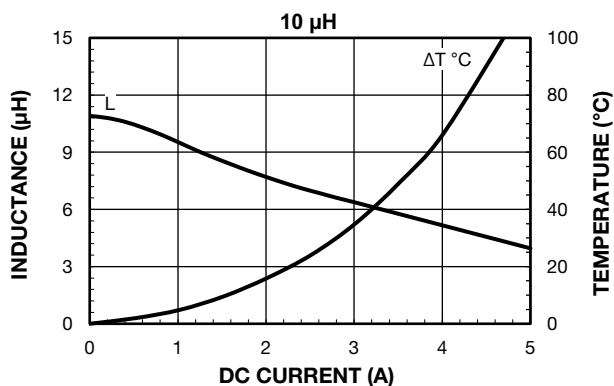
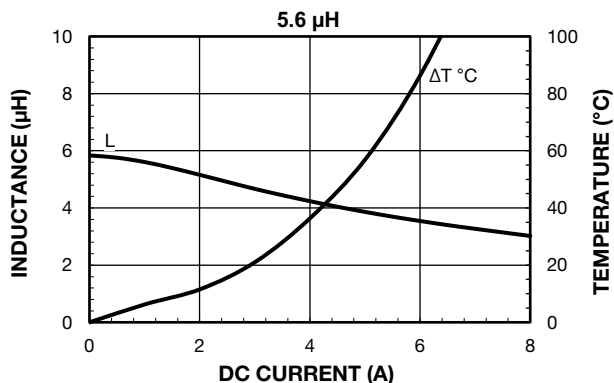
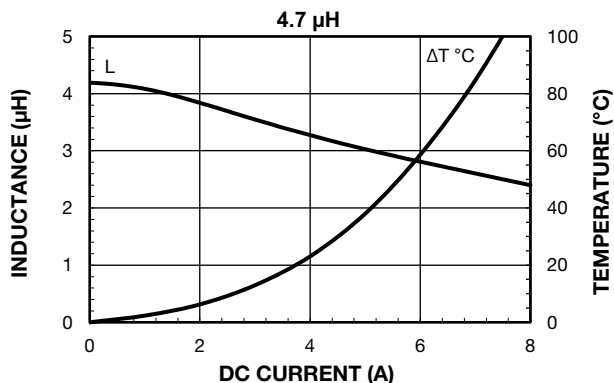


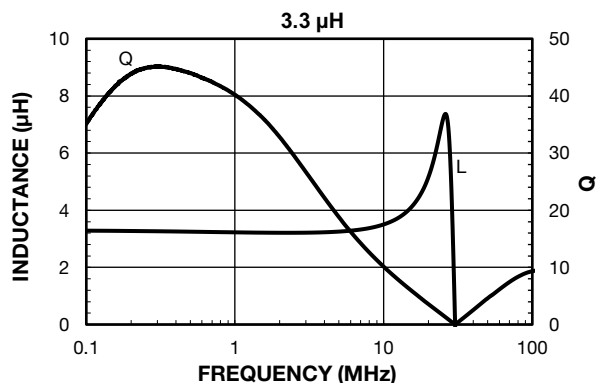
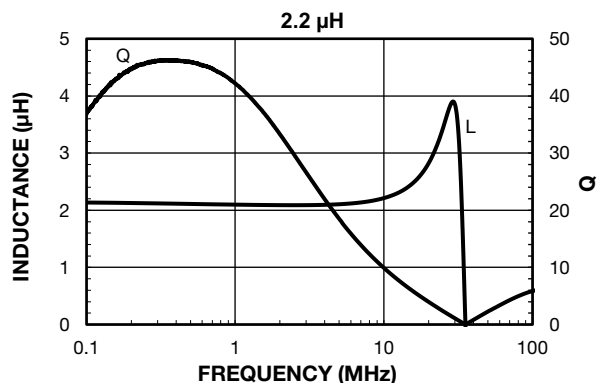
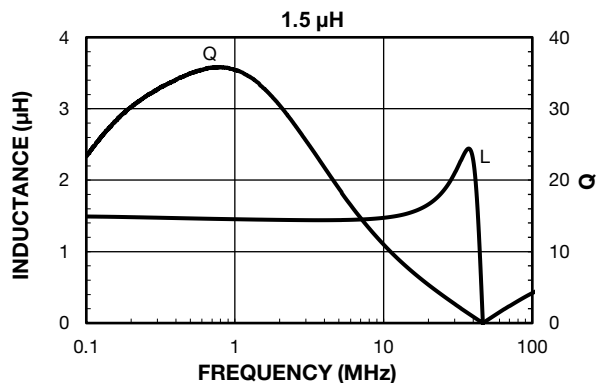
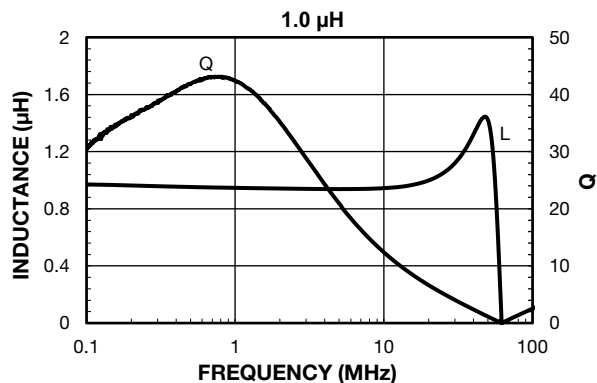
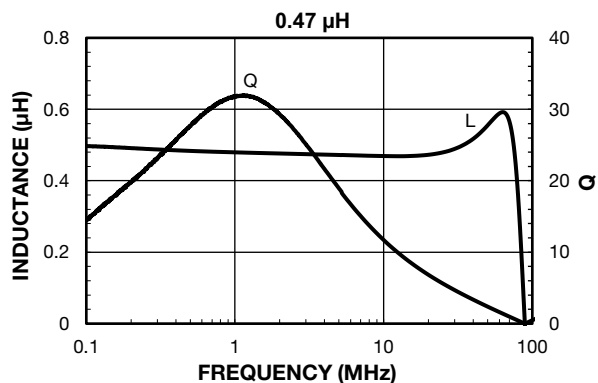
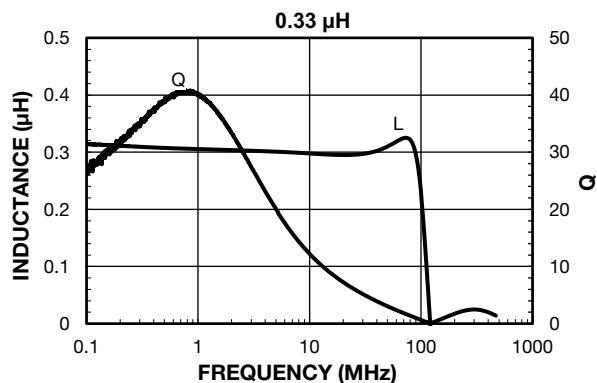
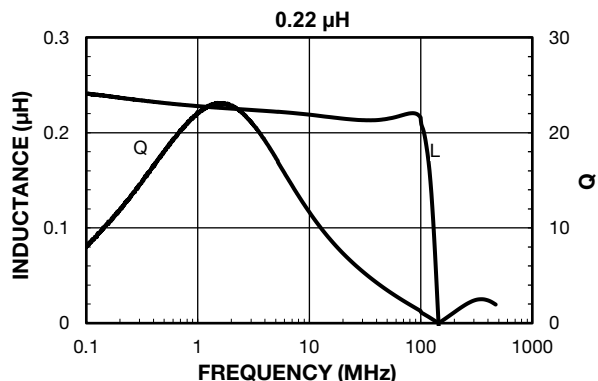
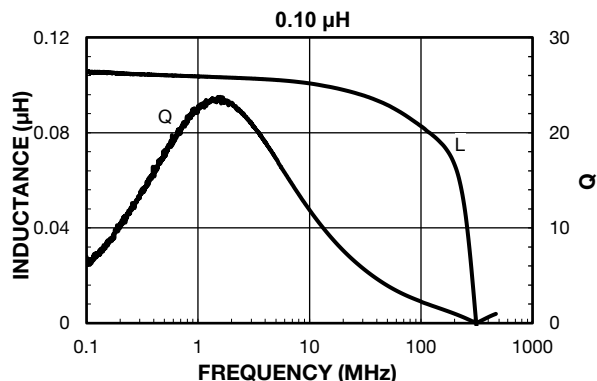
PERFORMANCE GRAPHS





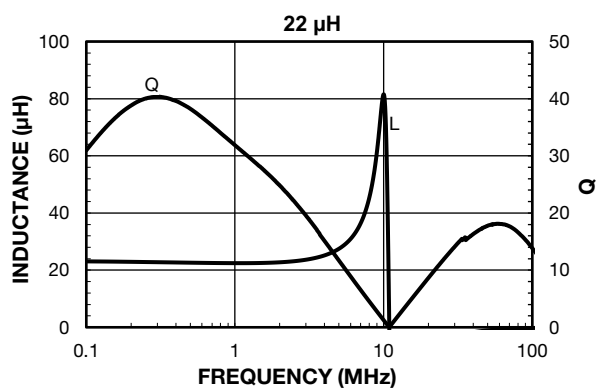
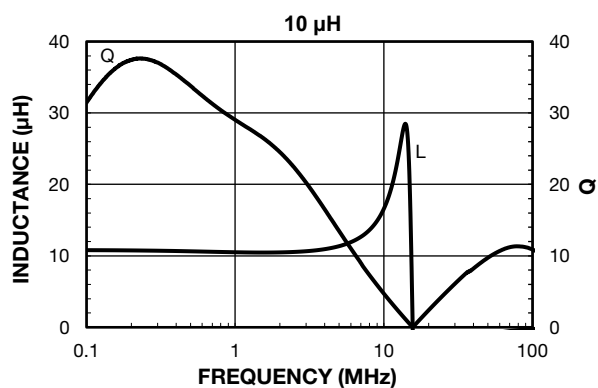
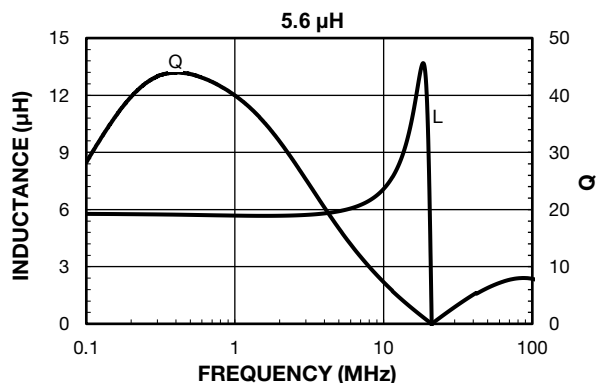
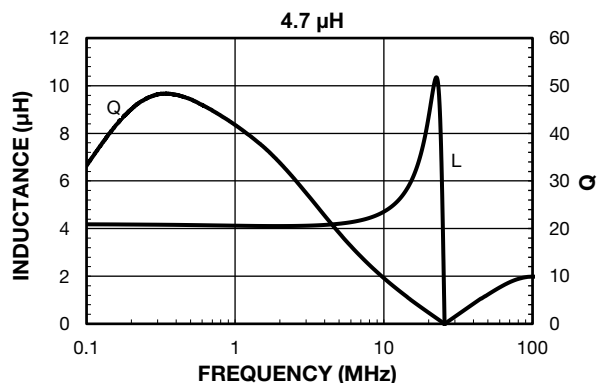
PERFORMANCE GRAPHS



PERFORMANCE GRAPHS: INDUCTANCE AND Q VS. FREQUENCY




PERFORMANCE GRAPHS: INDUCTANCE AND Q VS. FREQUENCY





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